DISCRETE MATHEMATICS

MODULE 1 – MATHEMATICAL LOGIC

Predicate Logic and Methods of Proof

Predicate Logic
Well-Formed Formula
Universal and Existential Quantifiers
Natural Deduction
Rules of Inference
Methods of Proof
Resolution Principle

Diploma – AI & ML

Lecture Notes

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1. Predicate Logic

1.1 Introduction

In propositional logic, we represent a complete statement using a proposition such as p or q.

For example,

$$p : \text{Ravi is a student}$$

This statement is either **True** or **False**.

However, propositional logic becomes less convenient when we want to describe **properties of objects** or make statements about **a group of objects**.

Consider the statement:

All students are hardworking.

This statement is not about one particular student. It refers to every student.

To represent such statements, we use **predicate logic**.

Definition

Predicate Logic is a form of logic used to represent statements about objects, their properties and relationships using variables and quantifiers.

Predicate logic extends propositional logic by introducing:

- Variables
- Predicates
- Quantifiers
- Domains

1.2 Predicate

A predicate describes a property of an object or a relationship between objects.

A predicate is commonly represented as

$$P(x)$$

where:

- x is a variable.
- P represents a property.

Example

Let

$$P(x) : x \text{ is an even number}$$

If $x = 4$,

$$P(4) : 4 \text{ is an even number}$$

which is True.
If $x = 5$,

$$P(5) : 5 \text{ is an even number}$$

which is False.

Therefore, the truth value of $P(x)$ depends on the value assigned to x.

1.3 Predicate versus Proposition

This is an important distinction.

Proposition	**Predicate**
Complete statement	Expression containing variable(s)
Has a definite truth value	Truth value depends on variable
Example: "5 is even"	Example: $P(x) : x$ is even

For example,

$$P(x) : x > 10$$

is a predicate.
But

$$P(15) : 15 > 10$$

is a proposition and is True.
Similarly,

$$P(5) : 5 > 10$$

is a proposition and is False.

Important Point

A predicate becomes a proposition when its variable is assigned a specific value or is properly quantified.

2. Universe of Discourse / Domain

When we use a variable, we must know what values the variable is allowed to take.
The collection of possible values is called the **Universe of Discourse** or **Domain**.

Definition

The **Universe of Discourse** is the set of all possible values that a variable can take.

It is commonly denoted by U.

Example

Suppose

$$U = \{1, 2, 3, 4, 5\}$$

and

$$P(x) : x > 3$$

The possible values of x are only

$$1, 2, 3, 4, 5.$$

x	$x > 3$
1	False
2	False
3	False
4	True
5	True

Therefore, the domain tells us where the variable gets its values from.

3. Quantifiers

A predicate such as

$$P(x) : x > 5$$

does not tell us whether the statement applies to all values or only some values.
Quantifiers allow us to specify this.
The two basic quantifiers are:

1. Universal Quantifier
2. Existential Quantifier

3.1 Universal Quantifier

The symbol for the universal quantifier is

$$\boxed{\forall}$$

It is read as:

- For all
- For every
- For each

Definition

The **Universal Quantifier** states that a predicate is true for every element in the specified domain.

The general form is

$$\boxed{\forall x\, P(x)}$$

Read as:

For every x, $P(x)$ is true.

3.2 Example of Universal Quantifier

Let

$$U = \{1, 2, 3, 4\}$$

and

$$P(x) : x < 5$$

Then

$$\forall x(x < 5)$$

means:

Every element of the domain is less than 5.

Check:

$$1 < 5 \quad \text{True}$$

$$2 < 5 \quad \text{True}$$

$$3 < 5 \quad \text{True}$$

$$4 < 5 \quad \text{True}$$

Therefore,

$$\boxed{\forall x(x < 5) \text{ is True}}$$

3.3 When is a Universal Statement False?

For a universal statement to be true, every possible case must be true.

Therefore, if even one case is false, the universal statement is false.

Example

Let

$$U = \{1, 2, 3, 4, 5\}$$

Consider

$$\forall x(x < 4)$$

Check:

$$1 < 4 \quad T$$

$$2 < 4 \quad T$$

$$3 < 4 \quad T$$

But

$$4 < 4 \quad F$$

Therefore,

$$\boxed{\forall x(x < 4) \text{ is False}}$$

Here $x = 4$ is a **counterexample**.

Important Point

To prove a universal statement false, it is enough to find one counterexample.

3.4 Existential Quantifier

The symbol for the existential quantifier is

$$\boxed{\exists}$$

It is read as:

- There exists
- For some
- For at least one

Definition

The **Existential Quantifier** states that a predicate is true for at least one element in the specified domain.

The general form is

$$\boxed{\exists x\, P(x)}$$

Read as:

There exists at least one x such that $P(x)$ is true.

3.5 Example of Existential Quantifier

Let

$$U = \{1, 2, 3, 4, 5\}$$

and

$$P(x) : x > 4$$

Then

$$\exists x(x > 4)$$

means:

There exists at least one number in the domain that is greater than 4.

Since

$$5 > 4$$

the statement is True.
Therefore,

$$\boxed{\exists x(x > 4) \text{ is True}}$$

3.6 When is an Existential Statement False?

An existential statement requires at least one successful case.
Therefore, it is false when no element in the domain satisfies the predicate.

Example

Let

$$U = \{1, 2, 3, 4, 5\}$$

Consider

$$\exists x(x > 10)$$

None of the elements in the domain is greater than 10.
Therefore,

$$\boxed{\exists x(x > 10) \text{ is False}}$$

3.7 Universal versus Existential Quantifier

Universal Quantifier	Existential Quantifier
$\forall$	$\exists$
For all	There exists
For every	For at least one
Every case must satisfy the predicate	Only one case needs to satisfy the predicate
One counterexample makes it false	False only when no case satisfies the predicate

Important Point

Easy way to remember:

$$\boxed{\forall = \text{EVERY}}$$

$$\boxed{\exists = \text{AT LEAST ONE}}$$

3.8 Practice with Quantifiers

Let

$$U = \{1, 2, 3, 4, 5\}.$$

Determine whether each statement is True or False.

1. $\forall x(x > 0)$
2. $\forall x(x < 5)$
3. $\exists x(x = 3)$
4. $\exists x(x > 10)$

Answers

1. True
2. False; counterexample is $x = 5$
3. True
4. False

4. Well-Formed Formula (WFF)

A logical expression must follow certain rules, just as a sentence in English must follow grammatical rules.

For example:

> Ravi is a student.

is a grammatically meaningful sentence.
But:

> Student Ravi is a.

does not form a proper sentence.
Similarly, logical expressions must follow the rules of logic.

Definition

A **Well-Formed Formula (WFF)** is a logically correct expression constructed according to the rules of the formal language of logic.

In simple words:

> A WFF is a correctly written logical expression.

4.1 Examples of WFF

The following are examples of well-formed formulas:

$$p \wedge q$$

$$p \rightarrow q$$

$$\neg p$$

$$P(x)$$

$$\forall x P(x)$$

$$\exists x P(x)$$

$$\forall x (P(x) \rightarrow Q(x))$$

4.2 Identifying a WFF

Consider

$$\forall x P(x)$$

This is a WFF.
Its components are:

- $\forall$ – quantifier
- x – variable
- $P(x)$ – predicate

Another example is

$$\forall x(P(x) \to Q(x)).$$

This is also a WFF.
It means:

For every x, if $P(x)$ is true, then $Q(x)$ is true.

5. Natural Deduction

Predicate logic is not only used to represent statements. We also use logic to derive conclusions from given information.

This process is called **deduction**.

Definition

Natural deduction is a method of proving a conclusion by starting with given premises and applying valid rules of inference step by step.

The basic idea is:

$$\boxed{\text{Premises} \to \text{Rules} \to \text{Conclusion}}$$

5.1 Example of Natural Deduction

Given:

$$p \to q$$

$$p$$

Therefore:

$$q$$

The natural deduction steps are:

Step	Statement	Reason
1	$p \to q$	Given
2	p	Given
3	q	Modus Ponens

Therefore, q follows logically from the premises.

6. Rules of Inference

Definition

A **Rule of Inference** is a valid pattern of reasoning that allows us to derive a new statement from given premises.

Rules of inference provide standard patterns that can be used to construct logical arguments.

6.1 Modus Ponens

The structure is:

$$p \to q$$

$$p$$

$$\therefore q$$

Example

If I study, I will pass.
I study.
Therefore, I will pass.

Here,

$$p = \text{I study}$$

and

$$q = \text{I pass.}$$

Therefore,

$$p \to q, \quad p \quad \therefore q$$

This is called **Modus Ponens**.

6.2 Modus Tollens

The structure is:

$$p \to q$$

$$\neg q$$

$$\therefore \neg p$$

Example

If it rains, the ground becomes wet.
The ground is not wet.
Therefore, it did not rain.

This is called **Modus Tollens**.

6.3 Hypothetical Syllogism

The structure is:

$$p \to q$$

$$q \to r$$

$$\therefore p \to r$$

Example

If I study, I understand the topic.
If I understand the topic, I perform well.
Therefore:
If I study, I perform well.

6.4 Disjunctive Syllogism

The structure is:

$$p \vee q$$

$$\neg p$$

$$\therefore q$$

Example

I will study Mathematics or Programming.
I will not study Mathematics.
Therefore, I will study Programming.

6.5 Summary of Important Rules

Rule	Premises	Conclusion
Modus Ponens	$p \to q,\ p$	q
Modus Tollens	$p \to q,\ \neg q$	$\neg p$
Hypothetical Syllogism	$p \to q,\ q \to r$	$p \to r$
Disjunctive Syllogism	$p \vee q,\ \neg p$	q

7. Methods of Proof

A **proof** is a logical argument used to establish that a mathematical statement is true.
Different situations require different methods of proof.
Important methods include:

1. Direct Proof
2. Proof by Contrapositive
3. Proof by Contradiction
4. Proof by Cases

7.1 Direct Proof

Definition

In a **direct proof**, we start from the given information and logically derive the required conclusion.

The basic structure is:

$$\boxed{\text{Given} \to \text{Logical reasoning} \to \text{Conclusion}}$$

Example

Prove: If n is even, then n^2 is even.

Since n is even,

$$n = 2k$$

for some integer k.
Squaring both sides:

$$n^2 = (2k)^2$$

$$n^2 = 4k^2$$

$$n^2 = 2(2k^2)$$

Therefore, n^2 is divisible by 2.
Hence,

$$\boxed{n^2 \text{ is even}}$$

7.2 Proof by Contrapositive

Suppose we want to prove:

$$p \to q.$$

Instead of proving it directly, we can prove its contrapositive:

$$\neg q \to \neg p.$$

We know that

$$\boxed{p \to q \equiv \neg q \to \neg p}$$

Therefore, proving the contrapositive proves the original implication.

Example

Original statement:

If n^2 is even, then n is even.

Contrapositive:

If n is not even, then n^2 is not even.

Since the original statement and its contrapositive are logically equivalent, either one can be used to establish the result.

7.3 Proof by Contradiction

Definition

In **proof by contradiction**, we assume that the statement to be proved is false and show that this assumption leads to a contradiction.

The basic structure is:

Assume the opposite

↓

Logical reasoning

↓

Contradiction

Therefore, the original assumption is false and the required statement is true.

Important Point

Simple idea:
Suppose we want to prove that a statement is true.
We temporarily assume that the statement is false.
If this assumption produces an impossible result or contradiction, then the assumption must be wrong.

7.4 Proof by Cases

Definition

In **proof by cases**, a problem is divided into different possible cases, and the conclusion is proved separately for each case.

For example, an integer is either:

- Even, or
- Odd.

Therefore, a proof involving all integers can sometimes be divided into two cases.

Case 1

n is even.

Case 2

n is odd.

If the required result is true in both cases, the result is established for all integers.

8. Resolution Principle

Resolution is another important method of logical inference.

Definition

The **Resolution Principle** is a rule of inference used to derive a new clause from two clauses containing complementary literals.

The basic form is:

$$(p \vee q)$$

$$(\neg p \vee r)$$

Therefore,

$$\boxed{q \vee r}$$

8.1 Understanding Resolution

Consider:

$$(p \vee q)$$

and

$$(\neg p \vee r)$$

The two expressions contain the complementary literals:

$$p$$

and

$$\neg p.$$

These complementary literals are eliminated.
The remaining literals are:

$$q$$

and

$$r.$$

Therefore,

$$q \vee r$$

is obtained.

Important Point

Easy method to remember resolution:

$$\boxed{\text{Find opposite literals} \rightarrow \text{eliminate them} \rightarrow \text{combine remaining literals}}$$

8.2 Example of Resolution

Given:

$$(A \vee B)$$

$$(\neg A \vee C)$$

The complementary literals are:

$$A, \neg A.$$

Remove them.
The remaining literals are:

$$B, C.$$

Therefore,

$$\boxed{B \vee C}$$

is the resolvent.

9. Practice Questions

A. Predicate Logic

1. What is a predicate?
2. What is the difference between a proposition and a predicate?
3. What is a Universe of Discourse?

B. Quantifiers

4. Write the symbol for the universal quantifier.
5. Write the symbol for the existential quantifier.
6. What does $\forall x P(x)$ mean?
7. What does $\exists x P(x)$ mean?

C. WFF

8. What is a Well-Formed Formula?
9. Identify whether the following is a WFF:

$$\forall x P(x)$$

D. Rules of Inference

10. Identify the rule:

$$p \to q$$

$$p$$

$$\therefore q$$

11. Identify the rule:

$$p \to q$$

$$\neg q$$

$$\therefore \neg p$$

E. Methods of Proof

12. What is direct proof?
13. What is proof by contradiction?
14. What is proof by contrapositive?

F. Resolution

15. Find the resolvent:

$$(p \vee q), (\neg p \vee r)$$

Answer

$$\boxed{q \vee r}$$

10. Quick Revision

Topic	Key Idea
Predicate Logic	Logic involving objects, properties and relationships.
Predicate	Property or relationship involving variables.
Domain	Set of possible values of a variable.
WFF	Correctly constructed logical expression.
$\forall$	For every / all.
$\exists$	There exists / at least one.
Natural Deduction	Deriving conclusions step by step from premises.
Rule of Inference	Valid pattern of reasoning.
Direct Proof	Given $\rightarrow$ reasoning $\rightarrow$ conclusion.
Contrapositive	Prove $\neg q \rightarrow \neg p$ instead of $p \rightarrow q$.
Contradiction	Assume the opposite and derive a contradiction.
Cases	Prove separately for each possible case.
Resolution	Eliminate complementary literals and combine the remaining literals.

11. Important Points to Remember

1. $P(x)$ is a **predicate**, not necessarily a proposition.
2. The **domain** tells us what values x can take.
3. $\forall$ means **every**.
4. $\exists$ means **at least one**.
5. One **counterexample** is enough to show a universal statement is false.
6. A WFF follows the rules of logical syntax.
7. Natural deduction derives conclusions from premises using valid rules.
8. Modus Ponens and Modus Tollens are important rules of inference.
9. In direct proof, start from the given information and reach the conclusion.
10. In proof by contradiction, assume the opposite and derive a contradiction.
11. In proof by contrapositive, prove $\neg q \rightarrow \neg p$ instead of $p \rightarrow q$.
12. In resolution, look for complementary literals such as p and $\neg p$.